Simulation of the collective behaviour of flocking sheep to a herding dog

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Introduction

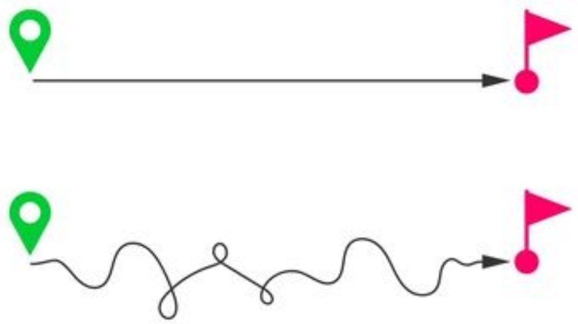
- Sheepherding
- Directional control
- Jadhav et al. (2024)
    - Our base model
    - Empirical study of sheep herded by a dog
    - Limitations

Why?

- Collective motion appears everywhere (animals, crows, traffic, swarms)
- Herding = controlled way to study group response to a threat
- Simple local rules -> complex group-level behaviour
- Practical application (games, robotics)

Starting Point & Goals

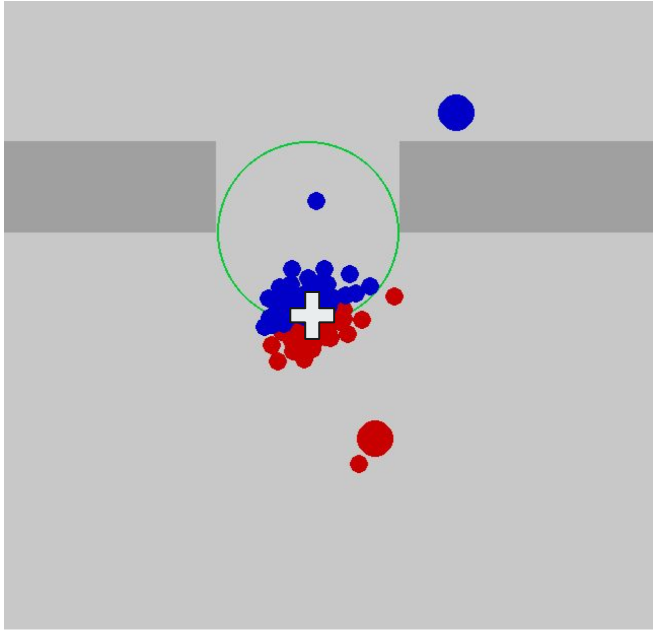
- Review herding simulations and the base model
- Reproduce base model results
- Visualization
- Extend the model:
  - Explicit target position
  - Environment obstacle
  - Two-dog herding
  - Dog obstacle avoidance?



Base Model Overview

Cohesion

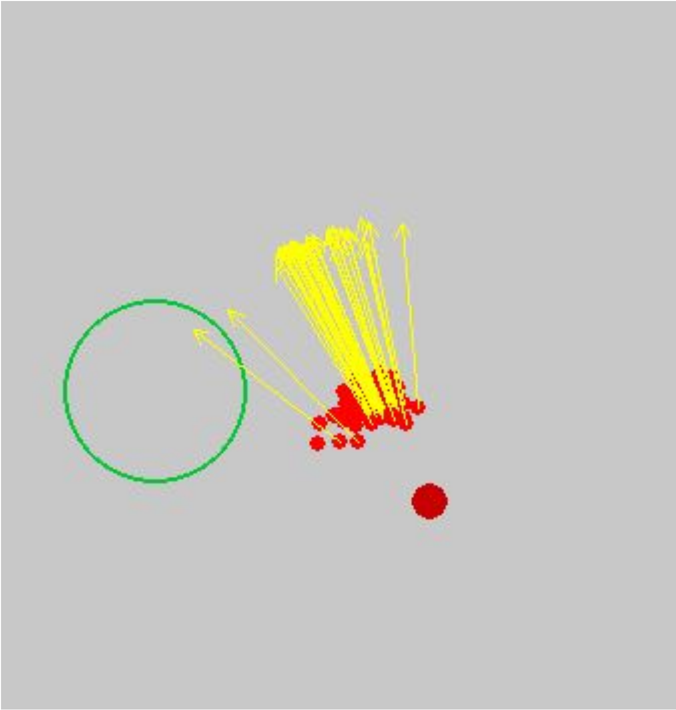
$$C(t) = \frac{1}{N} \sum_{i=1}^{N} d_{Bi},$$



Mean radius of the group - lower = better

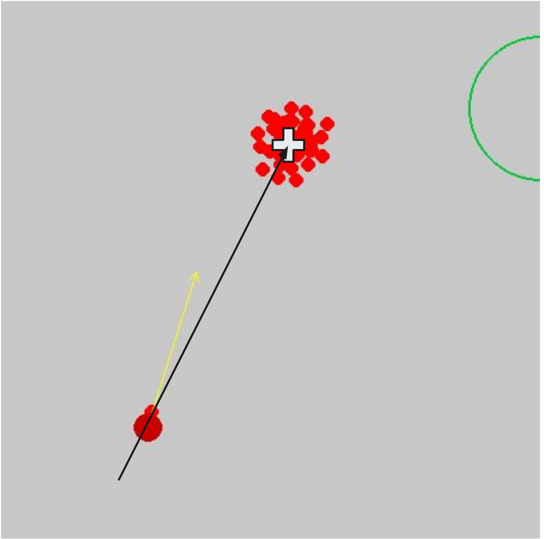
Polarization

$$P(t) = \frac{1}{N} \left\| \sum_{i=1}^{N} \frac{\vec{v}_i(t)}{\mathrm{v_i}(t)} \right\|$$

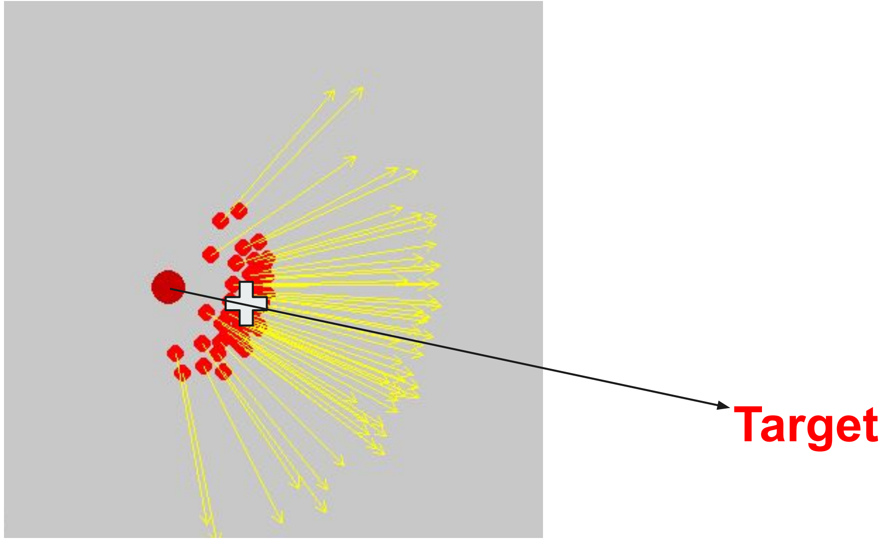


Degree of alignment of sheep

Collecting and driving



Collecting



Driving

Social sheep interactions

$$\vec{S}^{\,i}_{\mathrm{Att}} = \frac{w_{\mathrm{Att}}}{n_{\mathrm{Att}}} \sum_{j=1}^{n_{\mathrm{Att}}} \frac{\vec{r}_j - \vec{r}_i}{\|\vec{r}_j - \vec{r}_i\|}$$

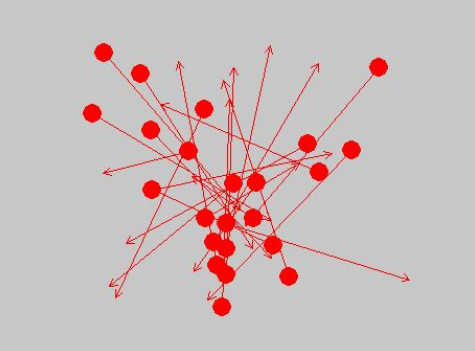
Social Attraction

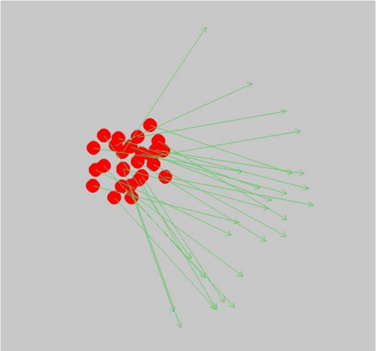
$$\vec{S}^{\,i}_{\mathrm{Ali}} = \frac{w_{\mathrm{Ali}}}{n_{\mathrm{Ali}}} \sum_{j=1}^{n_{\mathrm{Ali}}} \vec{e}_j$$

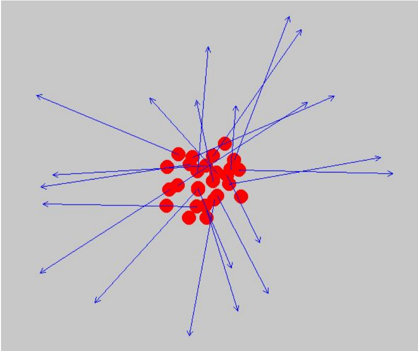
Social Alignment

$$\vec{S}^{\,i}_{\mathrm{Rep}} = -\frac{w_{\mathrm{Rep}}}{n_{\mathrm{Rep}}} \sum_{j=1}^{n_{\mathrm{Rep}}} \frac{\vec{r}_j - \vec{r}_i}{\|\vec{r}_j - \vec{r}_i\|}$$

Social Repulsion



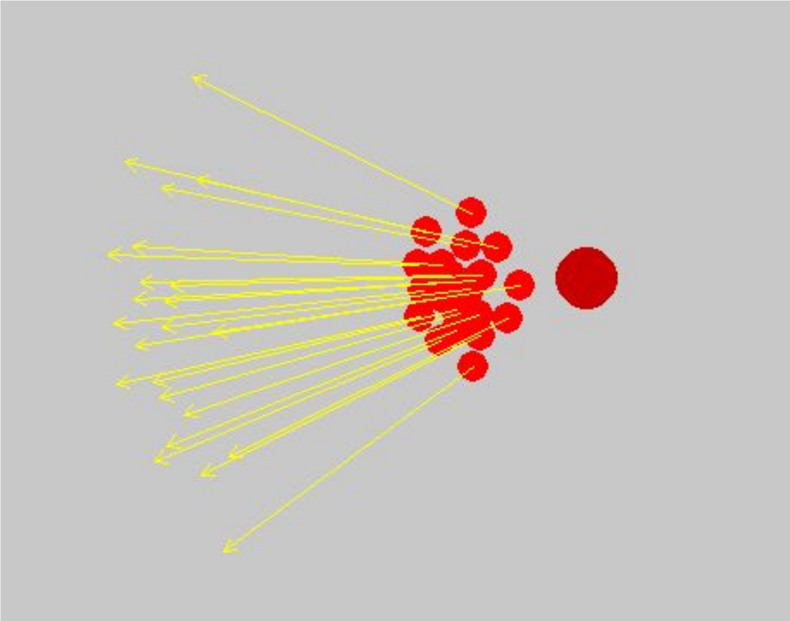




Dog - Sheep interaction

$$\vec{R}^{\,i}_{\text{dog}} = -w_{\text{dog}} \frac{\vec{r}_{\text{dog}} - \vec{r}_i}{\|\vec{r}_{\text{dog}} - \vec{r}_i\|}$$

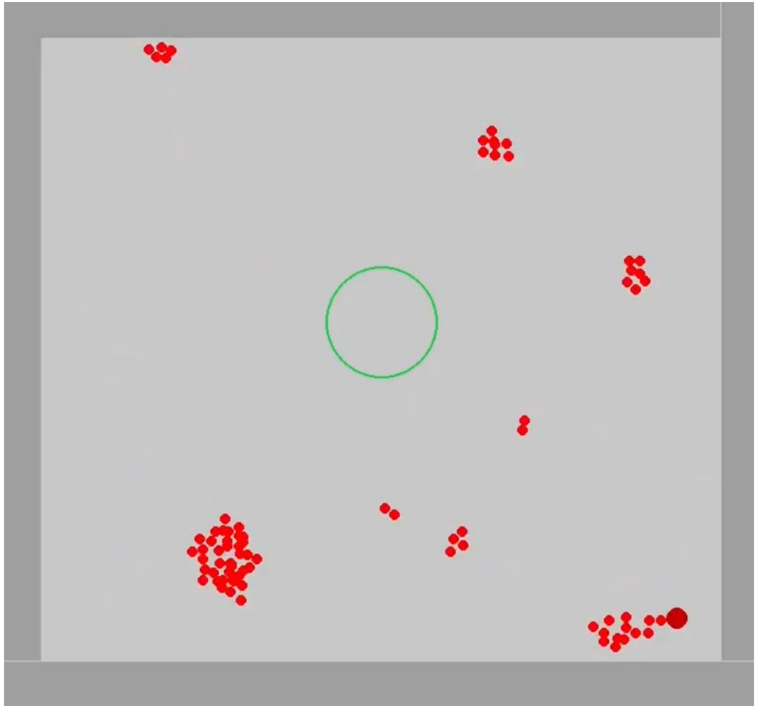
Dog Repulsion



Extensions

Vision limit

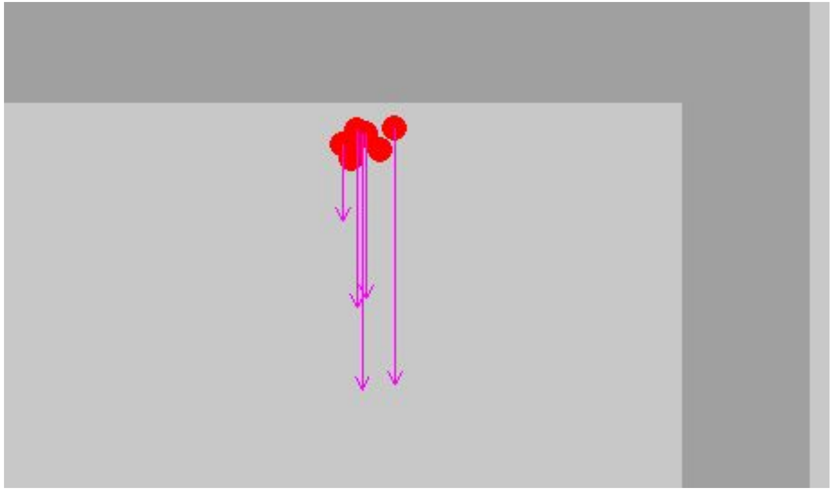
- Only considers neighbours within range
- Problem with larger areas
- Multiple independent flocks



Obstacles

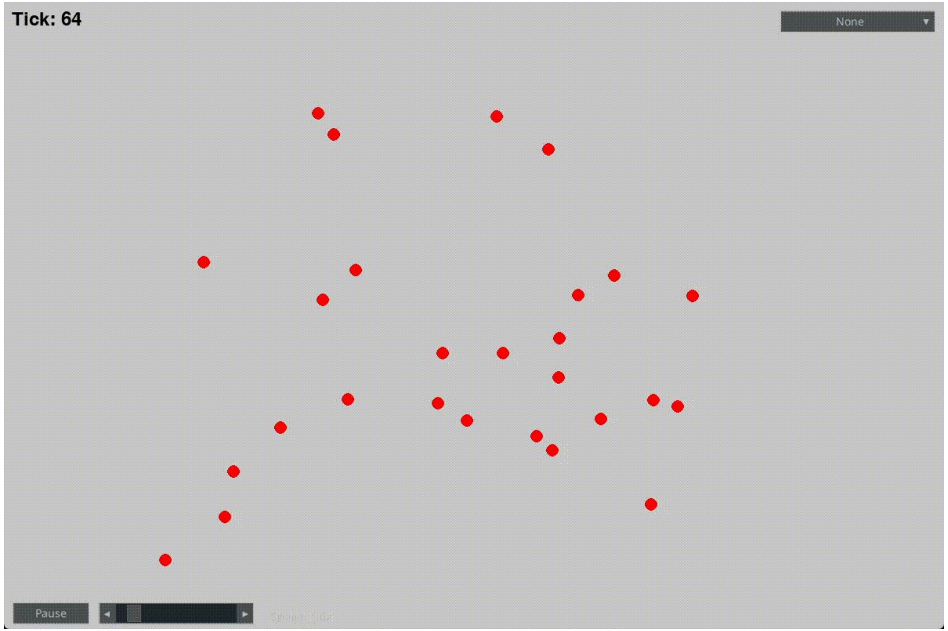
- AABB
- Normal of the surface

$$\overrightarrow{R}_{\text{obs}} = -w_{\text{obs}} \left(\frac{d_{\text{obs}} - d}{d_{\text{obs}}} \right)^2 \frac{\overrightarrow{d}}{\| \overrightarrow{d} \|}$$

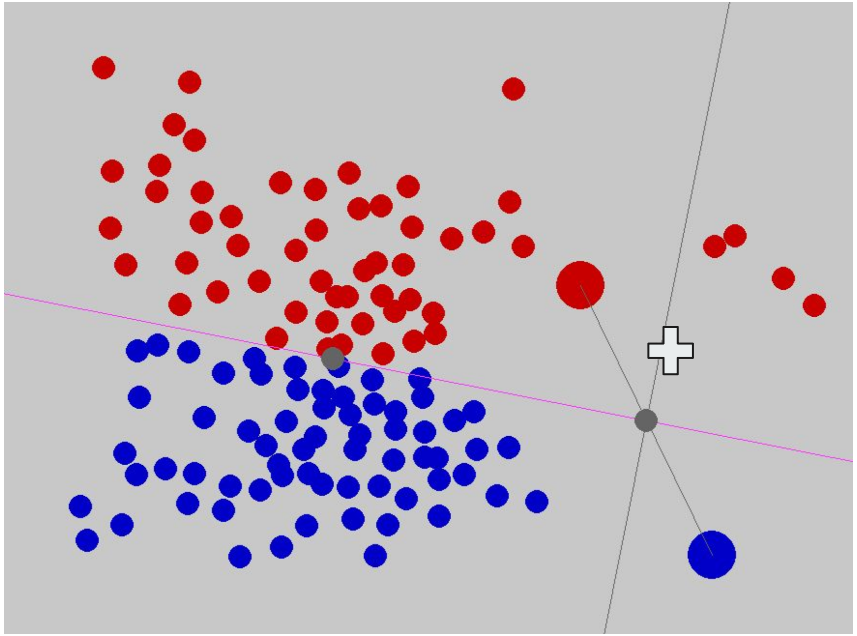


Idle state

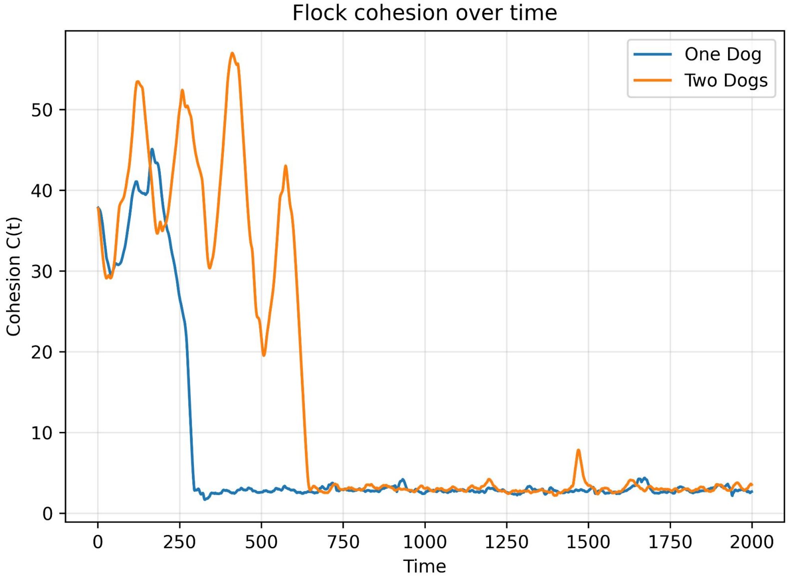
- Original model focuses on flock dynamics actively chased by the sheepdog
- We wanted grazing - collecting



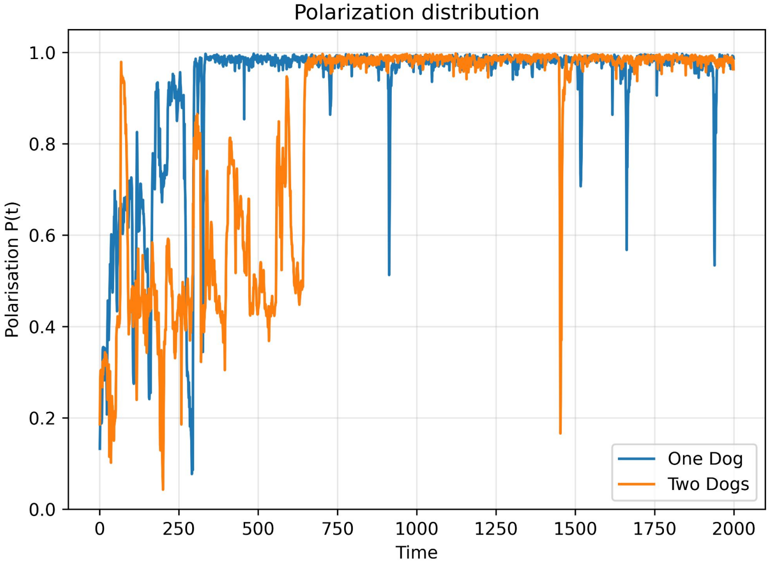
Two Dog Herding



Simulation



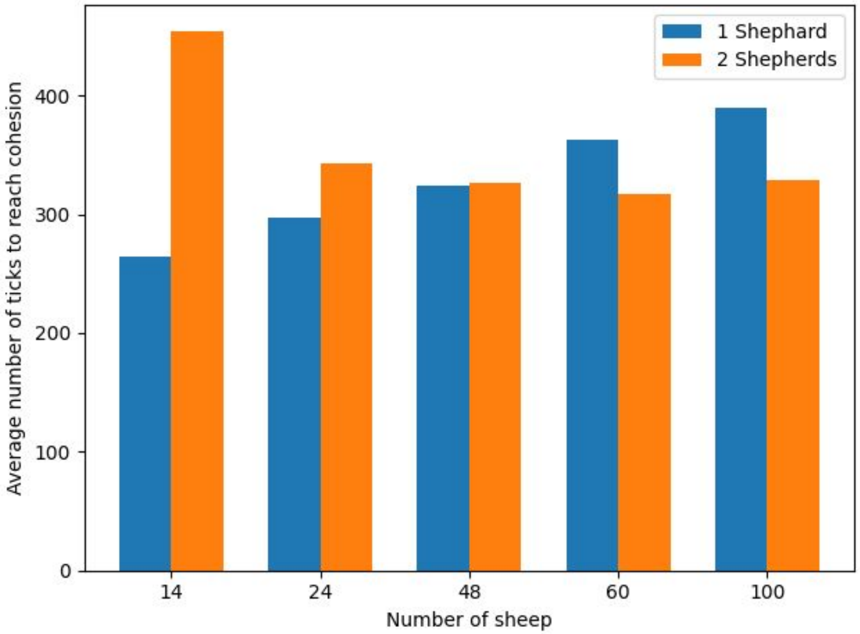
Flock cohesion over time
One Dog
Two Dogs
Cohesion C(t)
0
10
20
30
40
50
0
250
500
750
1000
1250
1500
1750
2000
Time



Polarization distribution
One Dog
Two Dogs
Polarisation P(t)
0.0
0.2
0.4
0.6
0.8
1.0
0
250
500
750
1000
1250
1500
1750
2000
Time

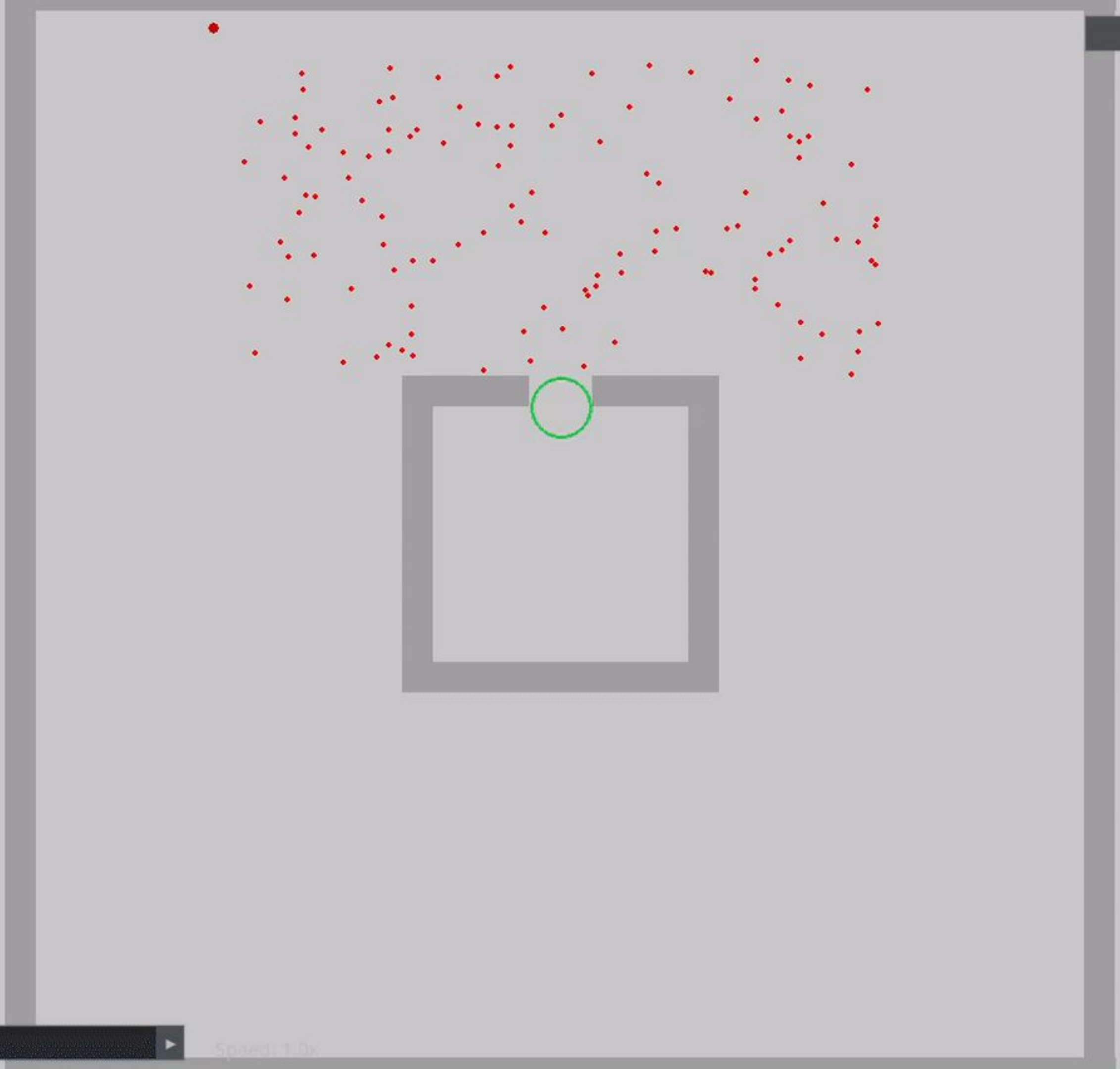
Simulation

- Multiple flock configurations
- How long to reach flock cohesion?
- Average of 20 runs



Video

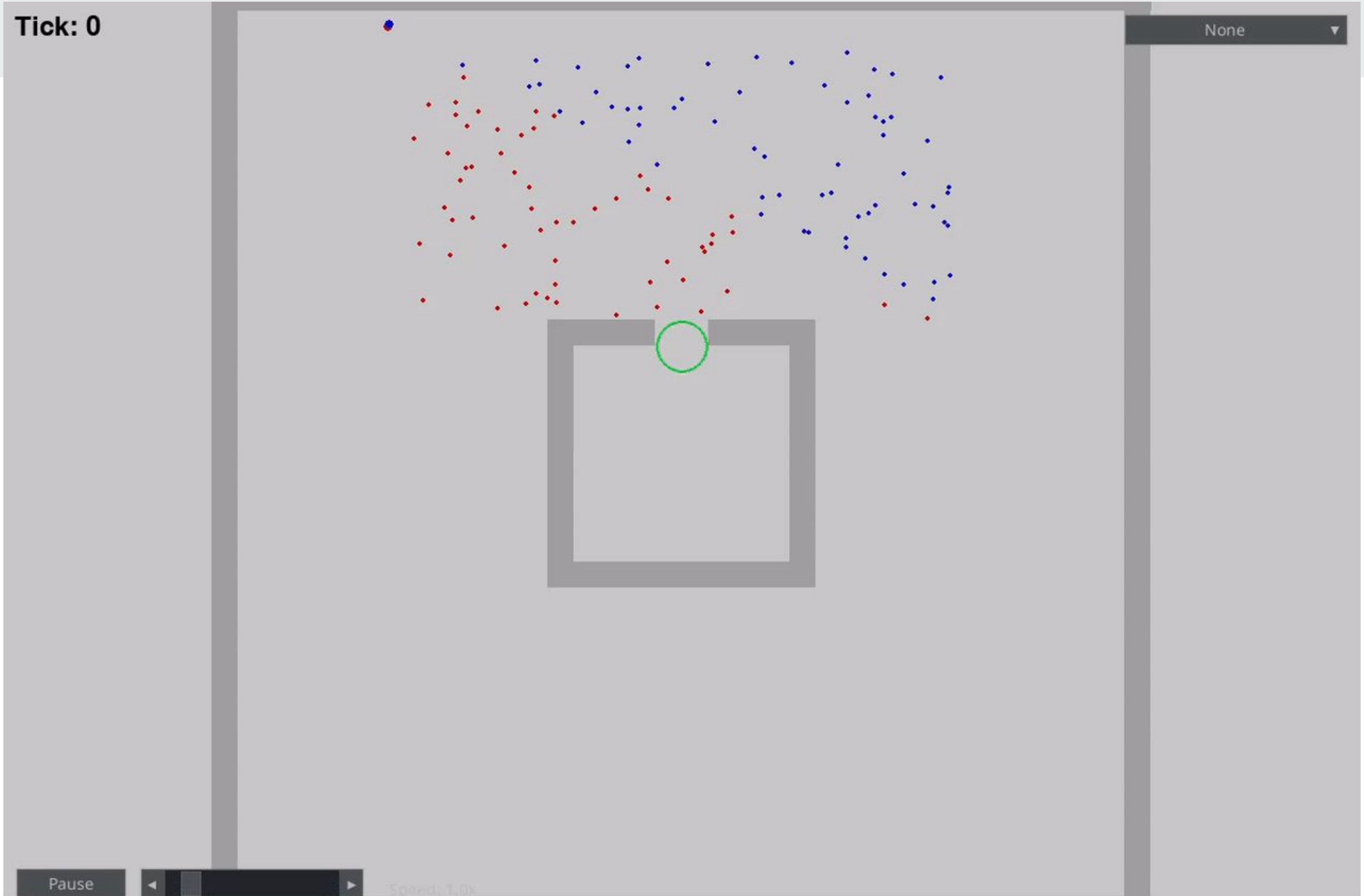
Tick: 0

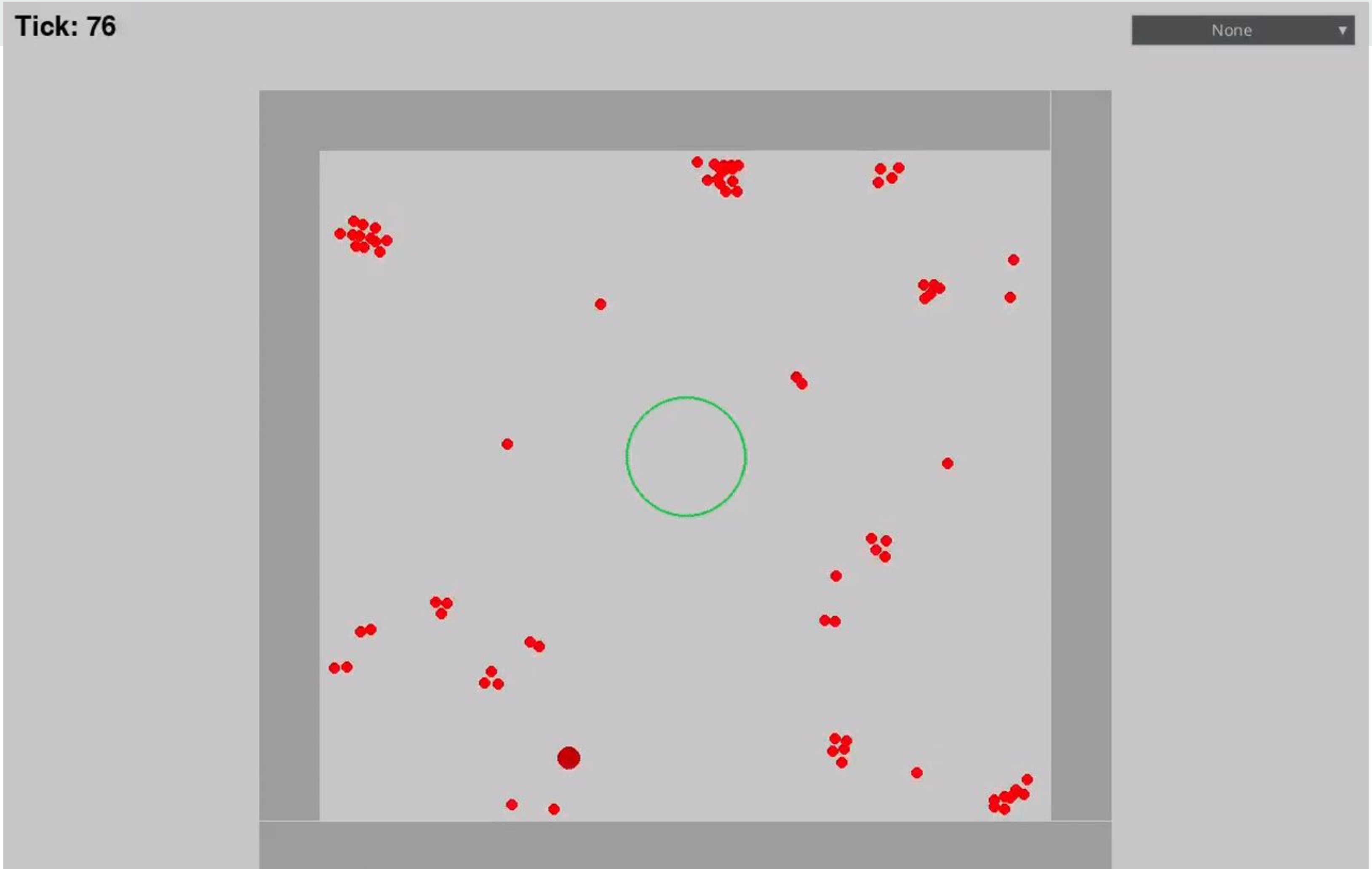


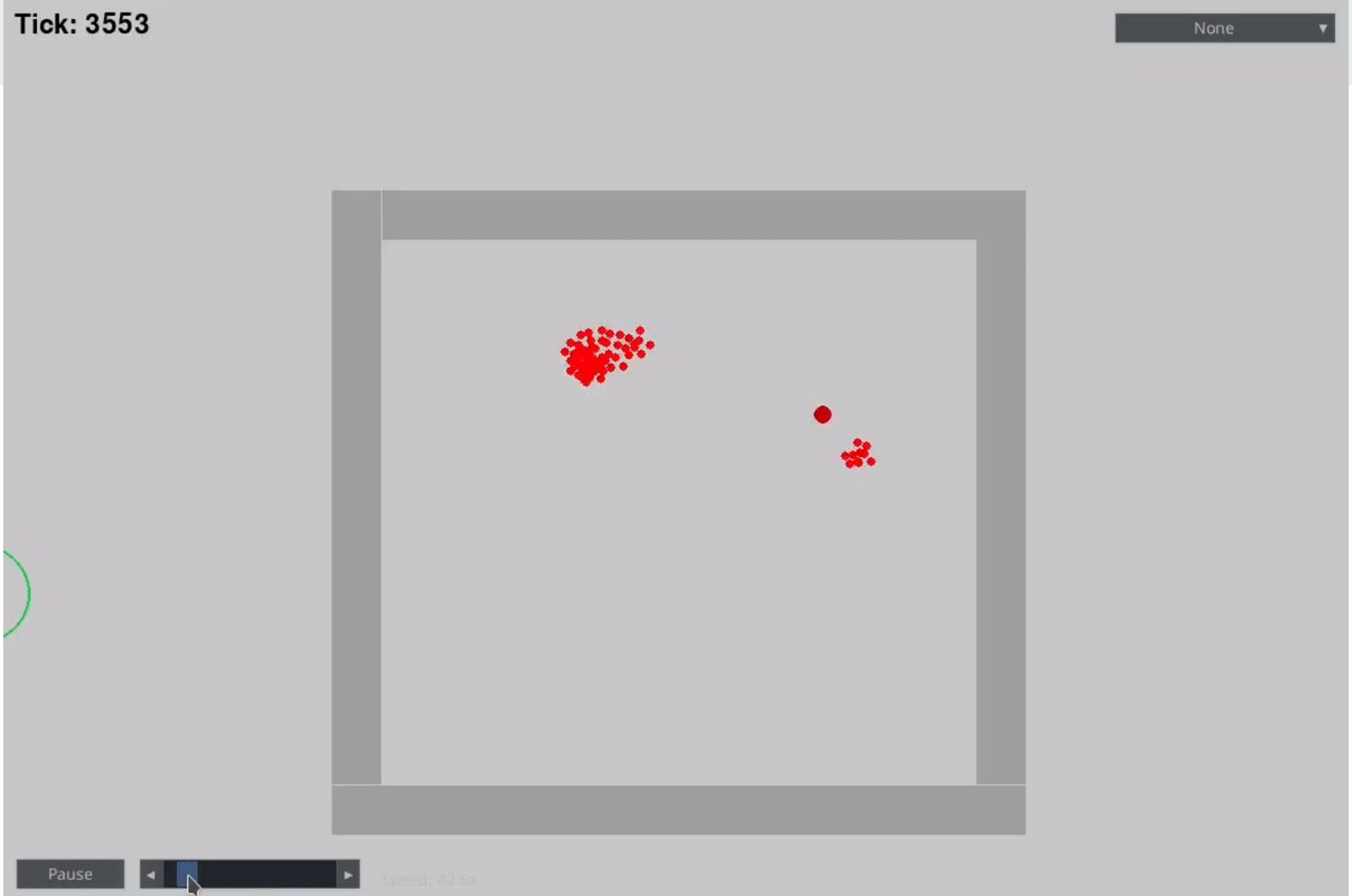
None ▾

Pause

Speed: 1.0x

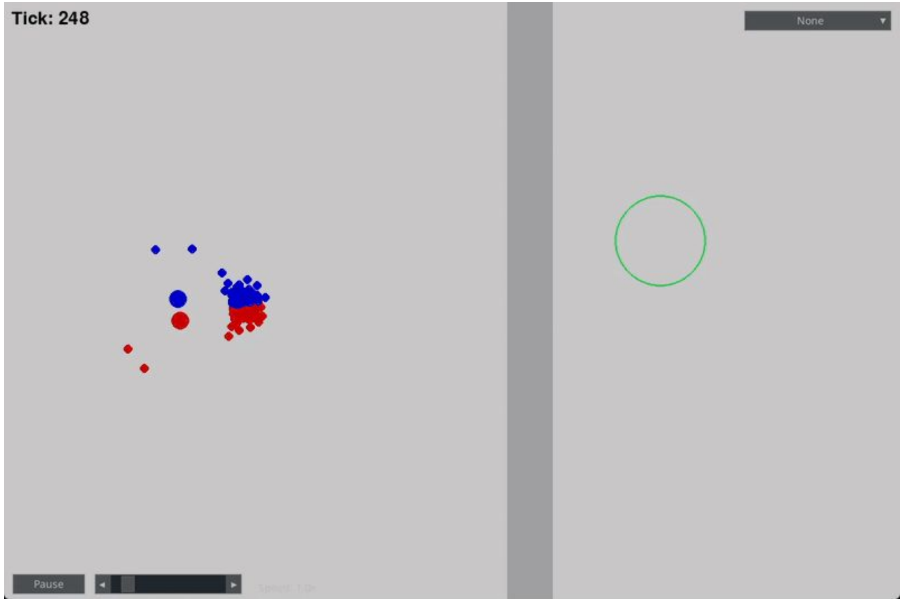

Tick: 0
None
Pause


Tick: 76
None


Tick: 3553
None
Pause

Further work

- Obstacles
- A more robust solution that redirects agents instead of repulse
- Would cause group to split
- More control agents != better performance



Conclusion

- We implemented the base algorithm Jadhav et al. (2024)
- Created a visualizer, several extensions and quality of life tools for easier testing
- Experimented with our model and explained the results
- Achieved variety of situations of flocking